Introduction to Cisco CLI

Quick notes on how the Cisco command line works, the modes, and the basic commands I use every day on switches and routers.

1. What is Cisco CLI

CLI means Command Line Interface. It is the text based way to talk to a Cisco device (switch, router, firewall). No mouse, no GUI. You type a command, press Enter, the device does it.

Cisco IOS is the operating system behind this CLI. Almost everything on the device can be checked and changed from here, so it is the first thing to learn.

2. How to connect to the CLI

- **Console:** direct cable from my PC to the console port. Works even when the device has no IP or the network is down. Used for first time setup and recovery.
- **SSH:** remote access over the network. Encrypted, so this is the one to use.
- **Telnet:** remote access but plain text, passwords go clear over the wire. Avoid it, disable it if possible.

Common tools: PuTTY, Tera Term, SecureCRT, or the normal ssh command from a terminal.

3. CLI modes

The CLI has different modes. What I can type depends on which mode I am in. The prompt tells me where I am, so always read the prompt first.

Prompt	Mode	What it is for
`Switch>`	User EXEC	Basic checking only. Very limited commands.
`Switch#`	Privileged EXEC	Full show commands, copy, reload, debug. Gateway to config mode.
`Switch(config)#`	Global config	Device wide settings like hostname, passwords, VLANs.
`Switch(config-if)#`	Interface config	Settings for one port or a range of ports.
`Switch(config-line)#`	Line config	Console and VTY (remote login) settings.

Moving between modes

```
Switch> enable                  ! go to privileged EXEC
Switch# configure terminal      ! go to global config
Switch(config)# interface gi0/1 ! go to interface config
Switch(config-if)# exit         ! go back one level
Switch(config-if)# end          ! jump straight back to privileged EXEC
Switch# disable                 ! back to user EXEC
```

Shortcut: Ctrl+Z does the same as end.

4. Getting help inside the CLI

- **?** shows all commands available in the current mode.
- **sh?** (no space) shows all commands starting with "sh".
- **show ?** (with space) shows what can come after "show".
- **Tab** completes the command so I don't type everything.
- **% Invalid input detected** means I typed something wrong. The ^ marker points to where the mistake is.
- **% Incomplete command** means the command needs more words. Use ? to see what is missing.

Commands can be shortened as long as they are not confusing. For example conf t works for configure terminal, and sh run works for show running-config.

5. Useful keyboard shortcuts

Key	What it does
`Up / Down arrow`	Go through previous commands
`Ctrl+A`	Move cursor to start of the line
`Ctrl+E`	Move cursor to end of the line
`Ctrl+C`	Cancel and exit config mode
`Ctrl+Z`	Exit config mode to privileged EXEC
`Ctrl+Shift+6`	Break out of a stuck command (ping, traceroute)

6. Basic show commands

These are the commands I use the most to check a device.

Command	What it shows
`show running-config`	Current config running in RAM
`show startup-config`	Saved config in NVRAM
`show version`	IOS version, uptime, model, memory
`show ip interface brief`	All interfaces with IP and up/down status
`show interfaces`	Detailed interface info, errors, speed
`show interfaces status`	Switch port status, VLAN, speed, duplex
`show vlan brief`	VLANs and which ports belong to them
`show mac address-table`	MAC addresses learned on switch ports
`show ip route`	Routing table
`show arp`	IP to MAC mapping
`show cdp neighbors`	Directly connected Cisco devices
`show logging`	Device log messages

Tip: add | include or | section to filter long output.

```
Switch# show running-config | include hostname
Switch# show running-config | section interface
Switch# show ip interface brief | include up
```

7. Basic configuration

Set hostname

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname SW-CORE-01
```

Set passwords

```
SW-CORE-01(config)# enable secret MyStrongPass
SW-CORE-01(config)# service password-encryption
```

Use enable secret, not enable password. The secret is stored hashed, the password is not.

Console line

```
SW-CORE-01(config)# line console 0
SW-CORE-01(config-line)# password ConsolePass
SW-CORE-01(config-line)# login
SW-CORE-01(config-line)# logging synchronous
SW-CORE-01(config-line)# exit
```

Login banner

```
SW-CORE-01(config)# banner motd # Authorized access only #
```

Configure an interface

```
SW-CORE-01(config)# interface vlan 1
SW-CORE-01(config-if)# ip address 192.168.1.10 255.255.255.0
SW-CORE-01(config-if)# no shutdown
SW-CORE-01(config-if)# description Management
SW-CORE-01(config-if)# exit
```

Enable SSH for remote access

```
SW-CORE-01(config)# ip domain-name mycompany.local
SW-CORE-01(config)# username admin secret AdminPass
SW-CORE-01(config)# crypto key generate rsa modulus 2048
SW-CORE-01(config)# ip ssh version 2
SW-CORE-01(config)# line vty 0 15
SW-CORE-01(config-line)# transport input ssh
SW-CORE-01(config-line)# login local
```

8. Save the configuration

Changes go to the running-config only. If the device reboots before I save, all changes are lost. Always save after the change is tested.

```
SW-CORE-01# copy running-config startup-config
SW-CORE-01# write memory          ! older short form, same result
```

9. Undo a command

To remove a setting, put no in front of the same command.

```
SW-CORE-01(config)# no banner motd
SW-CORE-01(config-if)# no ip address
SW-CORE-01(config-if)# shutdown       ! disable port
SW-CORE-01(config-if)# no shutdown    ! enable port
```

To go back to default for a setting, use default, for example default interface gi0/5.

10. Good habits

- Read the prompt before typing, so I know which mode I am in.
- Take a backup of the running-config before any big change.
- Use show commands to verify after every change.
- Give every interface a clear description.
- Never use Telnet. Use SSH with strong passwords.
- Do not save config until I confirm the change works.
- Be careful with shutdown on the port I am connected through, I can lock myself out.
- Use reload in <minutes> when doing risky remote changes, so the device comes back to the saved config if I lose access.

11. Quick cheat sheet

Command	Purpose
`enable`	Enter privileged EXEC
`configure terminal`	Enter global config
`interface gi0/1`	Select an interface
`show ip interface brief`	Fast interface status check
`show running-config`	See current config
`copy run start`	Save config
`no <command>`	Remove a setting
`end`	Back to privileged EXEC
`reload`	Restart the device